

Primary Quadratic Modules for Heisenberg Fixed Fibres

Route-A structural certificate C156

Abstract

For a frozen Heisenberg lattice automorphism we determine the Smith type of every horizontal fixed-class group, reduce the universal central-rotation denominator to its group exponent, and split the resulting quadratic function into orthogonal primary components. Exact component ledgers through iterate fourteen turn the clean fixed-circle count into a product of primary zero counts. Primary is finite-group terminology here, not an arithmetic factorization.

Keywords: Heisenberg nilmanifold; finite quadratic module; Smith normal form; primary decomposition; fixed fibres; exact verification.

中文摘要：本文研究一个固定的海森堡格自同构。我们先对每个迭代的水平不动类群给出史密斯标准型，再把中央旋转的统一分母降到该有限群的指数。所得二次函数可按有限阿贝尔群的主分量作正交分解，因此光滑不动圆周数等于各主分量零点数的乘积。直到第十四次迭代的精确枚举验证了结构定理及有限范围内的分母锐性；这里的主分量仅是群论概念，不代表算术局部因子。

关键词：海森堡幂零流形；有限二次模；史密斯标准型；主分解；不动纤维；精确验证。

1 Smith form and the exponent bound

Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, $B = A^n$, and $M = B - I$. With F_n, L_n denoting Fibonacci and Lucas numbers, the standard doubling and Cassini identities give

$$M = \begin{cases} L_n \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}, & n \text{ odd}, \\ F_n \begin{pmatrix} L_{n+1} & L_n \\ L_n & L_{n-1} \end{pmatrix}, & n \text{ even}. \end{cases} \quad (1)$$

The cofactors have determinants -1 and -5 , respectively; the latter has content one for even n . Consequently

$$G_n = \mathbb{Z}^2 / M\mathbb{Z}^2 \cong \begin{cases} (\mathbb{Z}/L_n\mathbb{Z})^2, & n \text{ odd}, \\ \mathbb{Z}/F_n\mathbb{Z} \times \mathbb{Z}/(5F_n)\mathbb{Z}, & n \text{ even}. \end{cases} \quad (2)$$

Its exponent is $h_n = L_n$ for odd n and $h_n = 5F_n$ for even n .

For $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the canonical lattice correction is

$$q_B(x, y) = \frac{ac}{2}x(x-1) + bcxy + \frac{bd}{2}y(y-1). \quad (3)$$

The polarization of (3) is $(Bv)_1(Bw)_2 - v_1w_2$. The actual iterate correction has the same polarization, so $q_n = q_{A^n} + \ell_n$ with ℓ_n integer linear. This drift cannot be dropped when zero levels are counted. For $m = Mv$, define $\rho_n([m]) = q_n(v) - m_1v_2 \pmod{1}$.

Here is the parity step behind the improved denominator. Write $M = gU$, $U = \begin{pmatrix} r & s \\ s & t \end{pmatrix}$, where $(g, \det U, h) = (L_n, -1, L_n)$ in the odd branch and $(F_n, -5, 5F_n)$ in the even branch. By (2), $W = hv = (X, Y)$ is integral. Formula (3) gives

$$N := 2h\{q_{A^n}(v) - m_1v_2\} = X(m_2 - ac) - Y(m_1 + bd) + hm_1m_2 \in \mathbb{Z}. \quad (4)$$

Modulo two, division by five in the even branch is harmless. Substitution of $B = I + gU$ reduces (4) to

$$N \equiv s(gr+1)(g+1)X + s(gt+1)(g+1)Y + \{g(rt+s^2) + r+t\}XY. \quad (5)$$

Both Fibonacci and Lucas numbers have parity period $0, 1, 1$. If $3 \mid n$, then $g = s = 0$ and $r = t = 1$; otherwise $g = 1$, $rt + s^2 = 1$, and $r + t = 1$. Thus (5) vanishes. Since $h\ell_n(v)$ is also integral, this proves

$$h_n\rho_n([m]) = 0 \quad \text{in } \mathbb{Q}/\mathbb{Z}. \quad (6)$$

This is an all-iterate divisibility result. Equality of the observed denominator with h_n is recorded only for $2 \leq n \leq 14$.

2 Orthogonal primary decomposition

If $u = Mw$, direct expansion gives the bilinear polarization

$$\beta_n([m], [u]) = v_1 u_2 - u_1 v_2 + m_1 u_2 \pmod{1}. \quad (7)$$

Thus primary components of G_n at distinct primes are orthogonal, and the restriction to a component of exponent p^e takes values in $p^{-e}\mathbb{Z}/\mathbb{Z}$. Uniqueness of primary torsion in $\mathbb{Q}/\mathbb{Z}$ gives

$$C_n = \prod_{p|h_n} C_{n,p}, \quad C_{n,p} = \frac{1}{p^e} \sum_{a \bmod p^e} \sum_{x \in G_{n,p}} e^{2\pi i a \rho_n(x)}. \quad (8)$$

Indeed, bilinearity kills cross-prime polarizations. The identity $\rho(kx) = k\rho(x) + \binom{k}{2}\beta(x, x)$ makes each odd-primary value p -primary; for $p = 2$, (6) removes the possible extra factor two. The formula is a finite root-of-unity zero projector, not an operator trace formula or an arithmetic local factor.

3 Exact certificate and boundary

CRT idempotents applied to the two standard generators, followed by column Hermite reduction, enumerate each primary component exactly. The counts are

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14
C_n	1	1	4	1	21	4	57	1	148	105	397	144	1041	57

For example, the 2-, 3-, and 5-primary zero counts at $n = 12$ are 16, 9, 1, while the 5-, 13-, and 29-primary counts at $n = 14$ are 1, 1, 57; their products give 144 and 57. The producer, direct-cocycle checker, and SymPy path independently verify the ledger. The strict tuple is (A1_FAIL, A2_FAIL, A3_FAIL, A4_FORMAL_HINT). Fixed sets remain clean circles and the ordinary isolated stability factor remains singular. We claim no isolated primitive-orbit determinant, target divisor, target functional equation or counting law, arithmetic local/Euler factor, root number, automorphy, Hilbert–Pólya construction, or Route-B authorization. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.

Declarations

Data availability. All claim-bearing data are the exact JSON evidence and scripts distributed in this C156 package. No external dataset is used. **Ethics.** The work uses no human participants, animals, or sensitive personal data. **Author contributions/CRedit.** This is an anonymous technical certificate with no individual authorship assignment; package provenance records the roles Conceptualization, Formal analysis, Software, Validation, and Writing. **Competing interests.** None known. **Funding.** No external funding is reported. **AI-use disclosure.** An AI coding assistant supported drafting, proof-check planning, and code review. Every quantitative statement is regenerated and independently checked by the packaged scripts; the assistant was not treated as an external peer reviewer.